

Cyclic dominance of a Potts–RPS model on complex networks

Full test and results report

Reproducible run 2026-07-02 · 18/18 steps OK · all figures + data regenerated byte-for-byte identical (deterministic seeds)

0. Model, notation and definitions

Agents sit on the nodes of a network and each holds one strategy in {Rock, Paper, Scissors} — a $q=3$ Potts spin. Interactions reward matching your neighbours (drives *order*) but also contain the cyclic RPS dominance (drives endless *cycling*); a single knob ε sets their ratio. The question throughout: for which $(\varepsilon, \text{network})$ does the population reach consensus, and when does it chase itself forever?

The model	
agent / node	one site of the network; holds strategy $s_i \in \{\text{R, P, S}\}$ and interacts only with its graph neighbours
payoff matrix	$P = I + \varepsilon \text{skew}$: the identity part pays for matching neighbours (ferromagnetic, orders); the antisymmetric part encodes the cycle <i>Paper beats Rock, Scissors beats Paper, Rock beats Scissors</i>
ε	cyclic-dominance strength ($0 \leq \varepsilon \leq 1$), the control parameter of every sweep
utility U_i	sum of P -payoffs of i 's strategy against each neighbour's; scales with degree k , so connectivity acts like an inverse temperature
Glauber update	a node adopts a proposed strategy with probability $1/(1 + e^{-\Delta U/T})$; temperature $T=0.65$ everywhere in this report
sweep, burn-in	one sweep = N attempted single-node updates (the MC time unit); burn-in = initial sweeps discarded before measuring
seed	RNG / graph-realisation seed; curves are averaged over seeds where stated
Networks	
ER	Erdős–Rényi random graph: edges placed independently; narrow, homogeneous degree distribution around $\langle k \rangle$
BA	Barabási–Albert graph: grown by preferential attachment; heavy-tailed $P(k)$ with <i>hubs</i> (nodes of very high degree)
$N, \langle k \rangle, P(k)$	number of nodes; average degree; degree distribution
Observables	
(r, p, s)	global fractions of the population playing Rock, Paper, Scissors
ψ	$r + p e^{i2\pi/3} + s e^{i4\pi/3}$: complex sum of the three fractions; large $ \psi \Leftrightarrow$ one strategy dominates
m_ψ	$ \langle \psi \rangle_t $, the time-averaged order parameter: $\rightarrow 1$ consensus (<i>ordered phase</i>), $\rightarrow 0$ when strategies chase each other (<i>cycling phase</i>)
ε_c	critical cyclic strength where order gives way to cycling; quoted at the interpolated $m_\psi=0.5$ crossing throughout
conversion	fraction of <i>free</i> (non-zealot) nodes playing the zealots' strategy
Methods	
MC	Monte Carlo: stochastic agent-level simulation on the actual graph — the ground truth
HMF	homogeneous mean field: three coupled ODEs for (r, p, s) in which every node feels the same mean degree $\langle k \rangle$
DMF	degree-based mean field: one (r, p, s) per degree class, coupled through $P(k)$ — resolves degree heterogeneity
RMSE	root-mean-square error of a mean-field $m_\psi(\varepsilon)$ curve against the MC curve over the sweep grid
FSS	finite-size scaling: how the transition sharpens and its apparent ε_c shifts as N grows — the signature of a true phase transition
Perturbations	
zealot	a node locked to one strategy forever: it influences neighbours but never updates; z = zealot fraction
defect	quenched (frozen-in) damage: a fraction f of edges removed (broken links) or nodes removed (vacancies)
effective $\langle k \rangle$	the mean degree of the network <i>after</i> damage

1. Run manifest and reproducibility

Every step below was produced in one clean run of `./run_all.sh` (clean-rebuild C++ engine $\rightarrow$ validation test $\rightarrow$ all simulations). Per-step logs live in `logs/`; the machine-readable manifest is `logs/manifest.csv`.

step	status	wall (s)
test_validate_engine	OK	4
mean_field_hmf	OK	1
mean_field_sweep	OK	7
mean_field_compare_suite_ER	OK	5
mean_field_compare_suite_BA	OK	6
monte_carlo_mc	OK	2
monte_carlo_compare	OK	77
phase_diagram_ER	OK	6
phase_diagram_BA	OK	6
phase_diagram_boundary	OK	0
dynamics_ternary	OK	2
dynamics_fss	OK	3
dynamics_stability	OK	1
zealots_experiment	OK	2
zealots_experiment_hubs	OK	4
zealots_experiment_mixed	OK	2
defects_experiment	OK	7
defects_collapse	OK	1
total	18/18 OK	136

Table 2: All simulation steps, status and wall-clock time (16-core host, C++20 engine `-O3 -march=native`). After the run `git status` reports zero changes to any tracked figure/CSV: fully deterministic.

2. Engine validation (C++ vs pure-Python MC)

What it is. A correctness cross-check: the same graph is evolved by the fast C++ engine and by an independent pure-Python Monte-Carlo reference, at four values of ε spanning both phases. Because the two use different RNG streams, agreement is expected in *regime* (ordered vs cycling), not in the last digit. This licenses using the C++ engine for everything below. Same graph through both engines; independent RNG streams, so agreement is in phase not last digit (log: `test_validate_engine.log`).

ε	Python m_ψ	C++ m_ψ	regime / verdict
0.2	0.999	0.999	agree (ordered)
0.5	0.991	0.992	agree (ordered)
0.7	0.004	0.007	agree (cycling)
0.9	0.004	0.003	agree (cycling)

Table 3: Engine agrees in every regime; measured speedup $31\times$ (Python 3.27 s vs C++ 0.10 s for the 4-point sweep; timing varies run-to-run, the m_ψ values do not).

3. Mean field and Monte Carlo

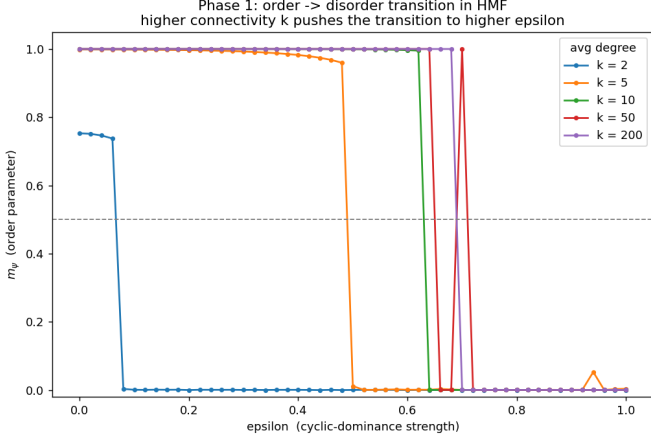
3.1 HMF ε -sweep: connectivity stabilises order

What it is. The homogeneous mean field replaces the network by three coupled ODEs for the population fractions (r, p, s) , in which every node feels the same mean degree $\langle k \rangle$. We integrate to steady state and sweep the cyclic-dominance strength ε at several $\langle k \rangle$, recording m_ψ . It is the cheapest probe of the order $\rightarrow$ cycling transi-

tion and how the critical point ε_c depends on connectivity.

$\langle k \rangle$	2	5	10	50	200
ε_c	0.08	0.50	0.64	0.66	0.70

ε_c rises monotonically with mean degree $\langle k \rangle$: denser networks sustain order against stronger cyclic pressure.



3.2 MC vs HMF vs DMF: RMSE against ground truth ($\langle k \rangle=10$)

What it is. A three-way accuracy test. We compute $m_\psi(\varepsilon)$ three ways on the same $\langle k \rangle=10$ setting: full agent-level **MC** (the ground truth), **HMF** (one mean degree), and **DMF** (degree-based mean field, one ODE class per degree, using the full $P(k)$). We score each mean field by its root-mean-square error against MC across the sweep. The question: does resolving the degree distribution (DMF) buy accuracy, and does that gain grow on a heterogeneous BA graph with hubs?

graph	RMSE(HMF)	RMSE(DMF)	DMF gain	$\varepsilon_c(\text{MC})$	$\varepsilon_c(\text{MF})$
ER	0.338	0.334	0.004	0.52	0.64
BA	0.336	0.328	0.008	0.52	0.64

Table 4: DMF beats HMF on both graphs; the DMF advantage is $\sim 2\times$ larger on heterogeneous BA. Both mean fields overestimate the ordered phase (MC turns over at $\varepsilon \approx 0.52$ vs ≈ 0.64).

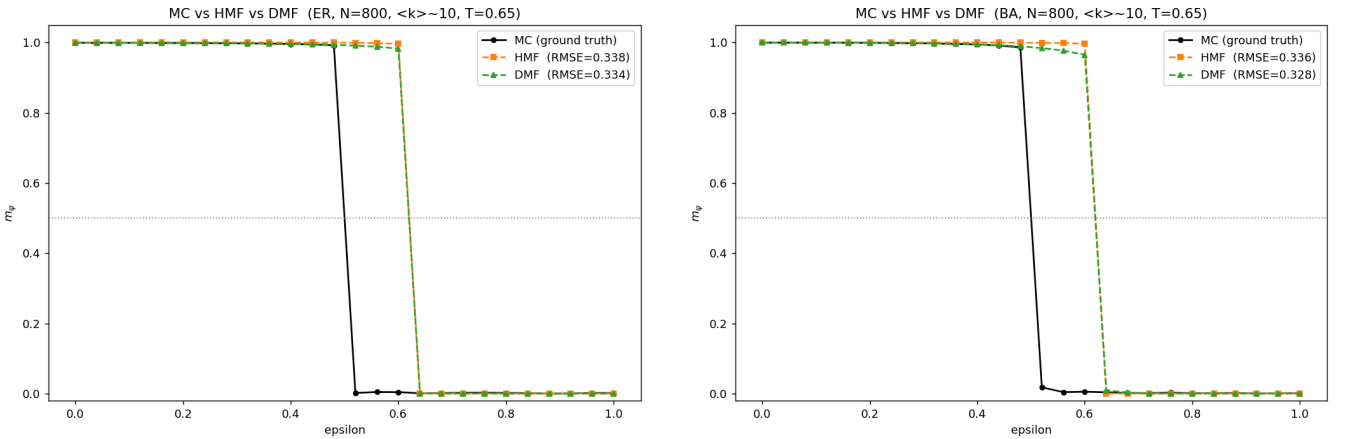


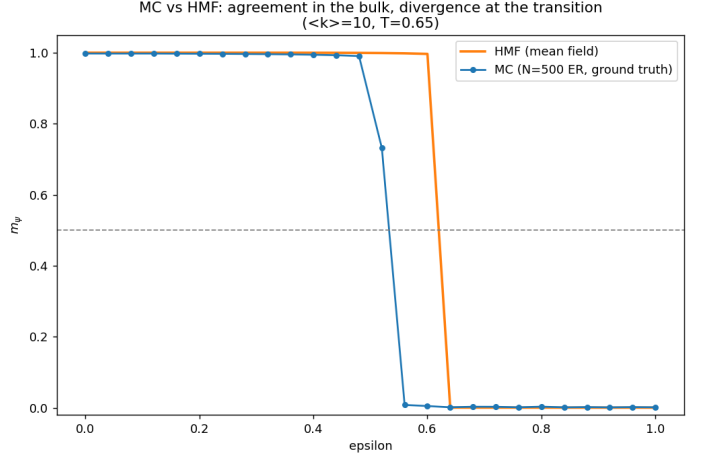
Figure 1: Comparison suite, ER (left) and BA (right).

3.3 Direct MC vs HMF overlay ($\text{ER}, \langle k \rangle=10$)

What it is. A head-to-head on a single ER graph: the MC transition curve overlaid on the HMF prediction, to see exactly where and by how much the mean-field approximation misplaces the critical point on a finite network.

model	ε_c
MC (ground truth)	0.56
HMF	0.64

MC transition precedes HMF: the mean field overestimates the ordered region, as expected on a finite network.



4. Phase diagram: $(\langle k \rangle \times \varepsilon)$ MC sweep

What it is. The central result of the study. For each mean degree we build a graph and run full MC across the whole ε range, tiling the $(\langle k \rangle, \varepsilon)$ plane with the measured order parameter to map the order/cycling boundary directly from agent-level dynamics (no mean-field assumption). Run for both ER and BA to isolate whether the boundary is set by average degree or by the shape of $P(k)$. 520 simulations per graph (20 degrees $\times$ 26 ε), $N=800$, fanned across 16 cores in ~ 6 s each.

$\langle k \rangle$	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36
ε_c (ER)	0.00	0.24	0.40	0.48	0.52	0.56	0.60	0.60	0.60	0.64	0.64	0.68	0.68	0.68	0.68	0.68	0.68	0.68
ε_c (BA)	0.00	0.24	0.40	0.48	0.52	0.56	0.60	0.60	0.60	0.64	0.64	0.68	0.68	0.68	0.68	0.68	0.72	0.72

Table 5: Transition ε_c vs mean degree for ER and BA. The order–cycling boundary bends upward with $\langle k \rangle$; ER and BA are nearly identical $\Rightarrow$ for MC dynamics average degree dominates over the shape of $P(k)$.

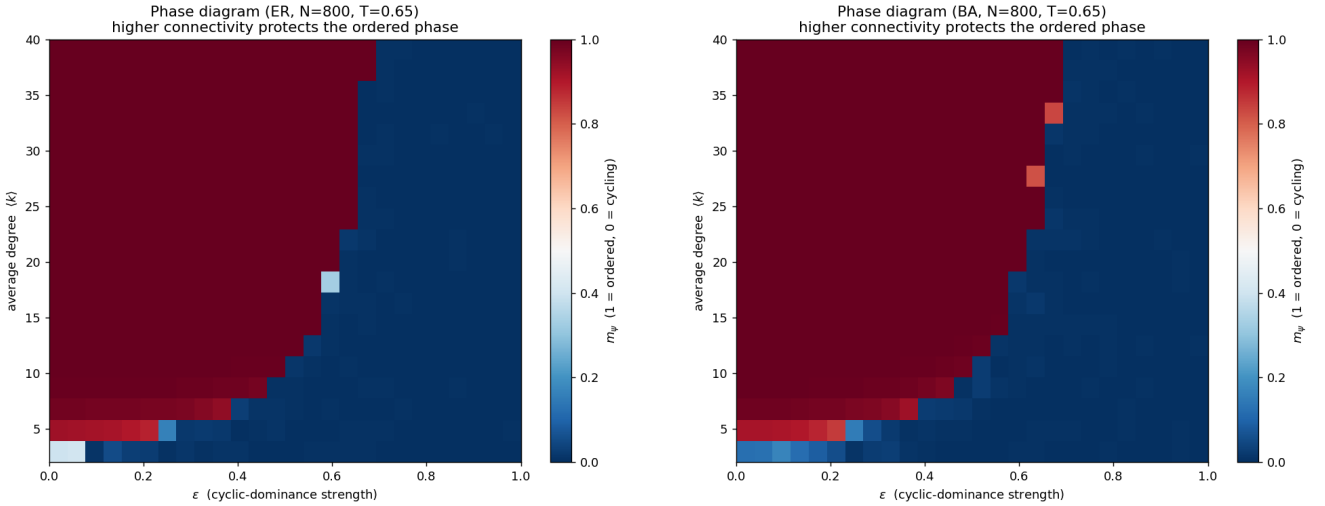


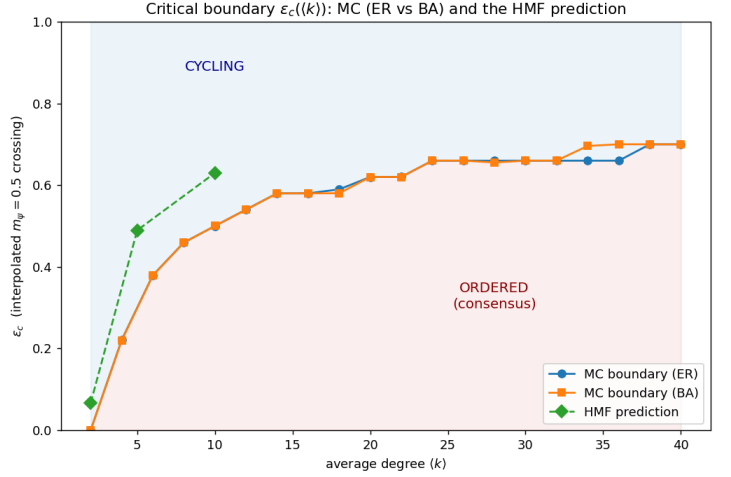
Figure 2: MC phase diagrams: ER (left), BA (right).

4.1 Extracted critical boundary $\varepsilon_c(\langle k \rangle)$

What it is. The boundary pulled out of both heatmaps as a curve (interpolated $m_\psi=0.5$ crossing per degree), with the HMF prediction from Sec. 3.1 overlaid. One figure carries three claims: the boundary rises with $\langle k \rangle$ (connectivity stabilises order), ER and BA coincide, and the mean field sits above the MC everywhere (overestimates order).

quantity	value
$\max_k \varepsilon_c^{ER} - \varepsilon_c^{BA} $	0.040
boundary range	0.00 $\rightarrow$ 0.70

The ER and BA boundaries agree within 0.04 at every one of the 20 degrees — the strongest quantitative form of “average degree matters, not $P(k)$ ”. Data: `critical_boundary.csv`.



5. Dynamics and finite-size scaling

What it is. Three views that confirm the transition is a real phase transition rather than a finite-size or dynamical artefact. **FSS:** rerun the MC transition curve at growing system size N and watch it sharpen and its midpoint converge. **Ternary portrait:** plot the trajectory in the (r, p, s) composition simplex — a fixed corner means consensus, a closed orbit means the endless RPS chase. **Stability:** classify the mean-field fixed points (consensus vs limit cycle) on either side of ε_c .

N	200	500	1000	2000
ε_c	0.570	0.530	0.510	0.510
$\max \text{slope} $	33	49	49	44

Table 6: Finite-size scaling (ER, $\langle k \rangle=10$): the transition sharpens and ε_c converges toward ≈ 0.51 as N grows 200 $\rightarrow$ 2000.

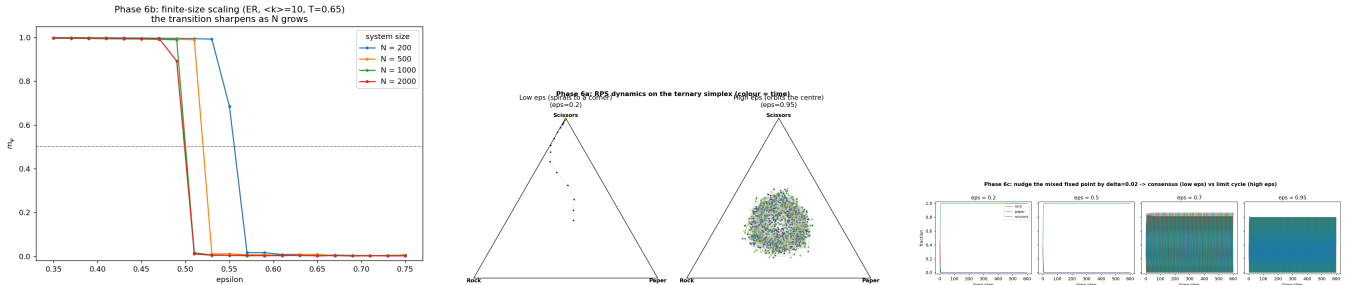


Figure 3: Left: FSS. Middle: ternary phase portrait (corner consensus below ε_c , limit cycle above). Right: fixed-point stability.

6. Perturbation experiments

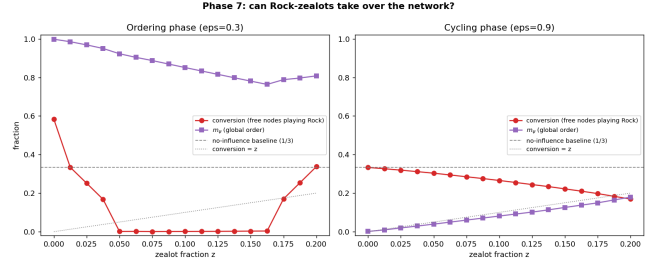
These studies build on the phase diagram: having mapped the clean transition, we now stress the ordered phase with targeted perturbations — stubborn agents, structural targeting, competition, and quenched network damage — to ask how robust consensus is and how a cyclic (RPS) system responds to interference. Each is run in both the ordering phase ($\varepsilon=0.3$) and the cycling phase ($\varepsilon=0.9$), averaged over many graph seeds.

6.1 Single Rock-zealot faction (ER, $\langle k \rangle=10$, 12 seeds)

What it is. A fraction z of nodes are turned into *zealots* permanently locked to Rock (they influence neighbours but never update). We grow z and track two things: *conversion* (what fraction of the remaining *free* nodes end up playing Rock) and the global order m_ψ . The classic question of whether a committed minority can drive consensus — but asked in a cyclic system, where pushing Rock also feeds Rock’s predator.

z	CONV_{ord}	$m_{\psi_{\text{ord}}}$	CONV_{cyc}	$m_{\psi_{\text{cyc}}}$
0.00	0.58	1.00	0.33	0.00
0.05	0.00	0.92	0.30	0.04
0.10	0.00	0.85	0.27	0.08
0.15	0.00	0.78	0.22	0.13
0.20	0.34	0.81	0.17	0.18

conv = fraction of *free* nodes playing Rock. Ordering phase: Rock-conversion collapses to ~ 0 by $z \approx 0.05$ — the free network flips to **Paper** (Rock’s predator). Cycling phase: only weak induced order, strategy cannot be pinned.

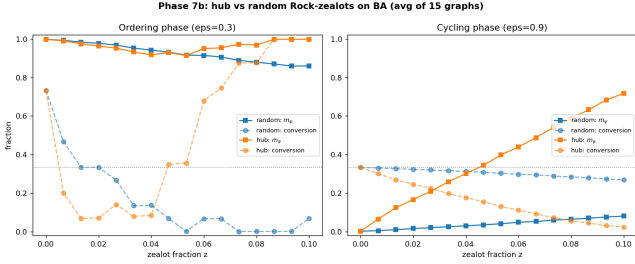


6.2 Hub vs random placement (BA, 15 seeds)

What it is. The same Rock-zealots, but now we compare *where* they sit on a scale-free BA network: spread at random versus concentrated on the highest-degree *hubs*. Since hubs touch far more neighbours, this tests how much raw structural targeting amplifies a minority’s reach — and whether that extra reach lets it dictate *which* strategy wins, or merely *whether* the network orders at all.

z	$m_{\psi_{\text{cyc}}}^{\text{rand}}$	$m_{\psi_{\text{cyc}}}^{\text{hub}}$	$m_{\psi_{\text{ord}}}^{\text{rand}}$	$m_{\psi_{\text{ord}}}^{\text{hub}}$
0.00	0.00	0.00	1.00	1.00
0.03	0.02	0.21	0.97	0.95
0.05	0.04	0.40	0.92	0.91
0.08	0.06	0.59	0.88	0.97
0.10	0.08	0.72	0.86	1.00

Cycling phase at $z=0.10$: hub $m_{\psi}=0.72$ vs random = 0.08 — a **$8.9\times$ amplification**. Hubs control *whether* the network orders, not *what* it orders on.

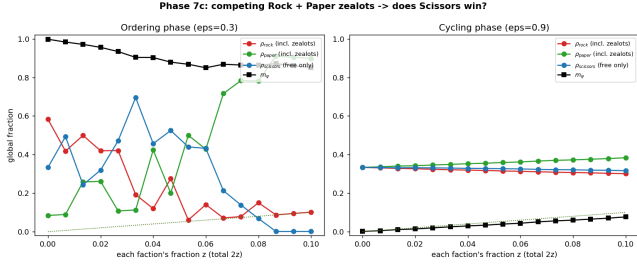


6.3 Competing Rock+Paper factions (ER, 12 seeds)

What it is. Two opposing zealot factions of equal size z — one locked to Rock, one to Paper — are grown together, and we watch the composition of the free population ($\rho_{\text{rock}}, \rho_{\text{paper}}, \rho_{\text{sciss}}$). In a cyclic game the naive guess is that the two cancel and the third strategy (Scissors, which beats Paper) profits; this experiment checks whether that actually happens.

z (each)	ρ_{rock}	ρ_{paper}	ρ_{sciss}	(ord.)
0.00	0.58	0.08	0.33	
0.03	0.42	0.11	0.47	
0.05	0.06	0.50	0.44	
0.08	0.15	0.78	0.07	
0.10	0.10	0.90	0.00	

With equal Rock+Paper zealots, the ordering-phase free population goes to **Paper** ($\rho \approx 0.90$), *not* Scissors: Paper is reinforced both by its own zealots and by the Rock-zealots it preys on. Cycling phase stays robust ($m_{\psi} \approx 0.08$).

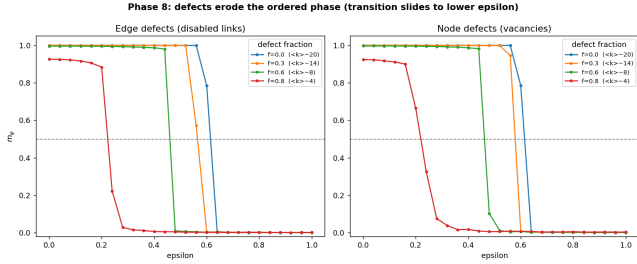


6.4 Network defects: edge vs node quenching (ER, 6 seeds)

What it is. Instead of adding agents we *damage* the network: remove a fraction f of edges (broken links) or of nodes (vacancies), then measure how the ε -transition moves. This is the mirror image of Sec. 3.1 — if connectivity stabilises order, quenched disorder should erode it — and comparing the two damage types (matched by the mean degree they leave behind) tests whether only the effective $\langle k \rangle$ matters.

f	$\varepsilon_c(\text{edge})$	$\langle k \rangle(\text{edge})$	$\varepsilon_c(\text{node})$	$\langle k \rangle(\text{node})$
0.00	0.64	20.0	0.64	20.0
0.30	0.60	14.0	0.60	14.1
0.60	0.48	8.0	0.48	8.0
0.80	0.24	4.1	0.24	4.0

Removing edges or nodes slides ε_c down ($0.64 \rightarrow 0.24$). Edge and node defects **coincide** once matched by the resulting $\langle k \rangle$ $\Rightarrow$ order-stability depends on effective $\langle k \rangle$ only.

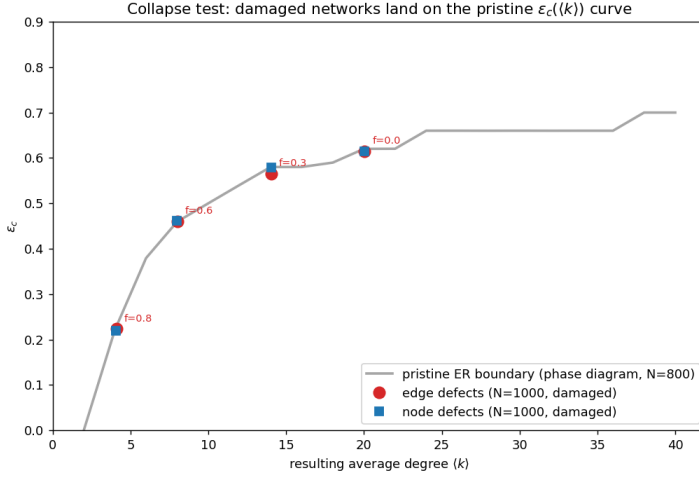


6.5 Collapse test: damaged $\equiv$ pristine at matched $\langle k \rangle$

What it is. The decisive check on Sec. 6.4: each damaged network contributes one point (resulting $\langle k \rangle$, ε_c), plotted over the *pristine*-ER boundary extracted in Sec. 4.1. If damage only matters through the mean degree it leaves behind, every point must land on the pristine curve.

quantity	value
$\max_f \varepsilon_c^{\text{edge}} - \varepsilon_c^{\text{node}} $	0.014
points on pristine curve	8/8

All eight damaged-network points (4 edge + 4 node, and at a different N than the pristine sweep) fall on the undamaged boundary: a damaged network is indistinguishable from a pristine one of the same average degree. Data: `defects_collapse.csv`.



7. Findings summary

1. **Connectivity stabilises order:** ε_c rises with $\langle k \rangle$ in HMF ($0.08 \rightarrow 0.70$) and in the MC phase diagram ($0 \rightarrow 0.72$).
2. **DMF > HMF:** lower RMSE vs MC on both graphs, advantage $\sim 2\times$ larger on heterogeneous BA. Both mean fields overestimate the ordered phase (MC $\varepsilon_c \approx 0.52$ vs MF ≈ 0.64).
3. **Average degree, not $P(k)$:** ER and BA phase boundaries coincide within $\max_k |\Delta \varepsilon_c| = 0.04$ over all 20 degrees.
4. **Genuine transition:** FSS sharpens and ε_c converges to ≈ 0.51 ; phase portrait shows corner consensus vs limit cycle.
5. **Zealots back-fire:** a Rock minority flips the free network to Paper (its predator), not Rock — naive minority takeover fails in a cyclic system.
6. **Hub leverage $\sim 9\times$:** hub-placed zealots amplify induced order vs random, but set *whether*, not *what*, the network orders on.
7. **Predator wins under competition:** equal Rock+Paper zealots drive the free population to Paper ($\rho \approx 0.90$), not Scissors.
8. **Defects $\equiv$ effective $\langle k \rangle$:** edge and node damage are equivalent once matched by resulting mean degree (gap ≤ 0.014); ε_c slides $0.64 \rightarrow 0.24$, and all damaged networks collapse onto the pristine boundary.